

Introduction to Data Management

- Practical Data Management for Animal Health Professionals
- Session 2A: Introduction to EpiCollect5
 - Tea at 10 – 1030 AM
 - Lunch 12-1300
 - Self session 1300 – 1500
 - Discussion at 1500
 - Self 1530-1630
 - Wrap at 1630
- Facilitator: John Grewar
- Date: 4 March 2025



Learning objectives

- Understand the basics of EpiCollect5 and its applications in field data collection
- Design and deploy data collection forms on mobile devices



What is Epicollect5?

- Epicollect5 is a mobile and web application for free and easy data collection
- It provides both web and mobile applications for the generation of forms (questionnaires) and freely hosted project websites for data collection
- Projects are created by using the web application at five.epicollect.net and then downloaded to the device to perform the data collection
- Data are collected (including GPS and media) using multiple devices and all data can be viewed on a central server (via map, tables, and charts)
- Data can be exported in **csv** and **JSON** format
- The mobile app is currently available for both Android (10+) and iOS (15+)



*Centre for Genomic Pathogen Surveillance. 2025. Epicollect5.
Available at: <https://five.epicollect.net>*



Projects and Entries Syncing

- Projects you make are available on searching on your mobile when logged in to the same account or if you have access to an already created project
- Data is stored on your mobile and is required to be manually uploaded to the server when you have internet access (i.e. hybrid storage with manual syncing)



Privacy

- Projects are public or private (preferred)
- Public projects mean anyone can see your data
- Private projects are only available to the people that you have given access to – we will only deal with private projects
- Users sign in with their Google, Apple or generic email accounts
- Privacy policy of EpiCollect is available at <https://docs.epicollect.net/about/privacy-policy>



Web application

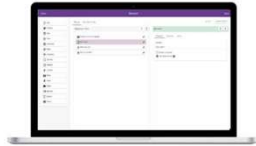
<https://five.epicollect.net/>



Hi, John My Projects Create Project Find Project User Guide Logout

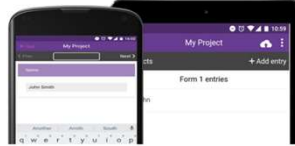
Free and easy-to-use mobile data-gathering platform.

Available on  



Create your project and forms

[TELL ME MORE](#)



Collect data online or offline

[TELL ME MORE](#)



View, analyse and export your data

[TELL ME MORE](#)

Have a look at our featured projects below or [FIND A PROJECT](#)



Features

- Drag and drop form builder
- Single form or multi forms survey (up to 5 hierarchy forms as child forms)
- Photo, video and audio recording
- Barcode scanner
- GPS location
- Branches (Dynamic lists)
- Groups (More questions on the same page)
- Conditional logic (jump questions)
- View data on a table or map
- Download data (CSV or JSON)
- Project cloning and sharing
- Data mapping to fit existing third party systems
- Export data via API
- User roles management



Project

- Project
 - Form(s)
 - Entries
- Project settings
 - Collaboration and access



- EpiCollect contains multiple Projects for each user.
- These projects can contain a parent form with either CHILD forms or BRANCHES that allow multiple hierarchies of data to be captured.

Create a project

<https://docs.epicollect.net/web-application/create-a-project>



Hi, John

My Projects

+ Create Project

Find Project

User Guide

Logout



Equine Holding
CENSUS FORM

FARM INFORMATION

FARM NAME		X
SURNAME		Y
NAME		
INITIALS		ADDRESS
TELEPHONE		
EMAIL		
CELL		
PRIVATE VET		
DATE OF CENSUS	/	/

HORSE INFORMATION

Name/ID	Passport/Chip	Sex	Birth date (if only year is known then choose 1 Aug of that year)	Colour	Breed	Species	Birth date (if only year is known then choose 1 Aug)	Birth date (if only year is known then choose 1 Aug)	Sample ID if applicable



- Start by creating a project in your profile. In this example we are replicating the Equine Census form used for outbreak investigation

Forms

Western Cape Government
Agriculture

Equine Holding CENSUS FORM

FARM INFORMATION

FARM NAME: _____ X _____

SURNAME: _____ Y _____

NAME: _____

INITIALS: _____

TELEPHONE: _____

EMAIL: _____

CELL: _____

PRIVATE VET: _____

DATE OF CENSUS: ____/____/____

HORSE INFORMATION

Name/ID	Passport/Chip	Sex	DOB (if only year is known then show 1 Aug of that year)	Colour	Breed	Species	and date 1 Date if known then show 1 Aug	and date 2 Date if known then show 1 Aug	Sample ID if applicable

New Project Import Project

Project name
dataMx_example
Max 50 chars

Small description (A long description can be added later)
An example project for teaching purposes
Max 100 chars

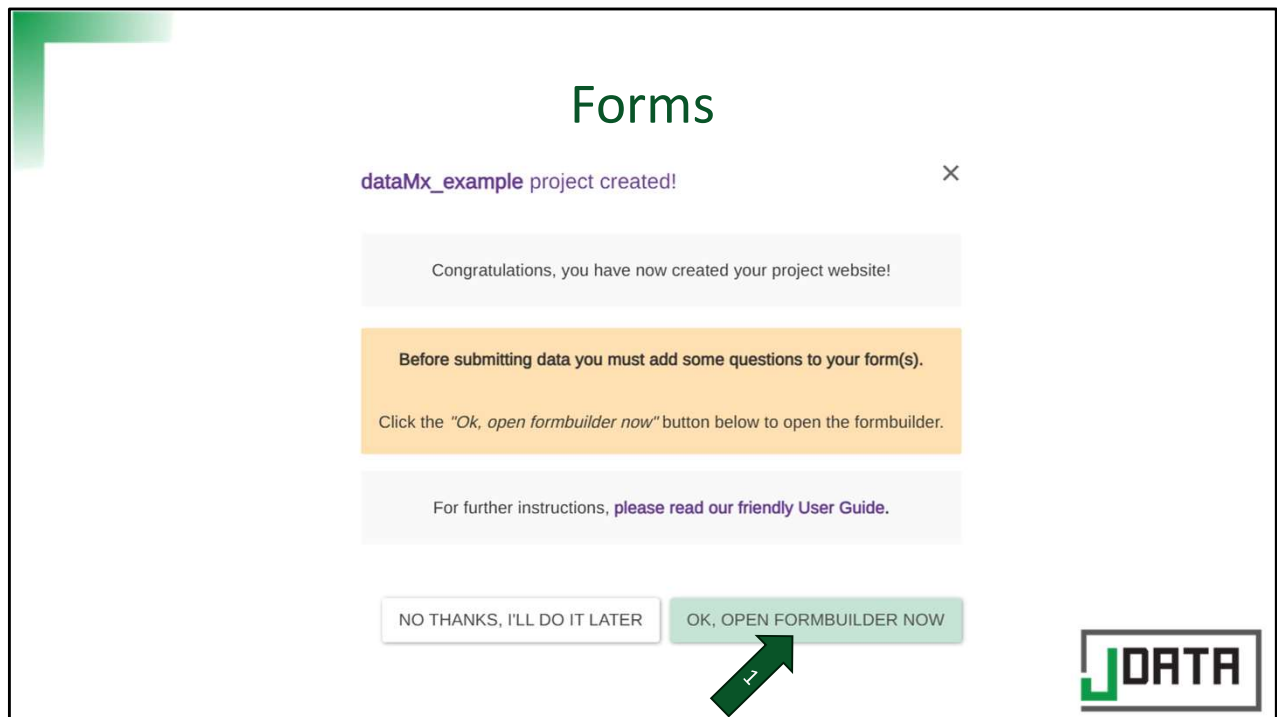
Form name
farminformation
Max 50 chars

Access
☒ Private ☐ Public

CREATE

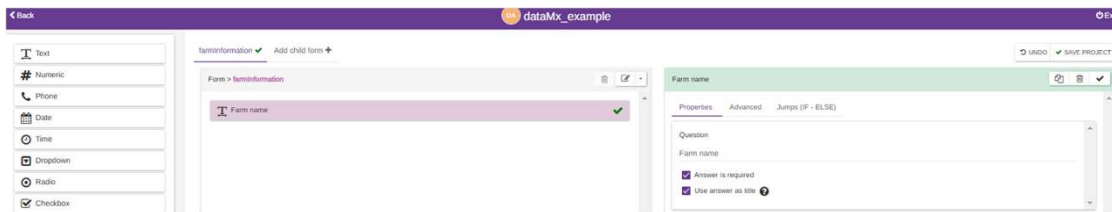
JDATA

- Allocate a project name, a short description and then the main parent form name.
- In this case our DataMx_example project has a farminformation form.
- The user access is set here to Private
- Click CREATE



- Go ahead and open the form builder

Forms



The form builder has three sections

- * Left column: the available inputs
- * Middle column: the inputs added to each form
- * Right column: the currently selected input settings

* By default, each input is shown on a single screen on the mobile app. This allows space for the popup keyboard and the question at the same time. You might want to have more than one question on a page, in that case just use a group – see <https://docs.epicollect.net/formbuilder/groups>

*We will now build the Equine form

Notes on data types

- PHONE: A question that allows only numbers, will use a phone keyboard layout on the mobile device. It accepts answers up to 255 chars.
- If multiple options are appropriate use a CHECKBOX type
- TEXT BOX: A big box to enter text on multiple lines. It accepts answers up to 1000 chars
- The format of DATE can be set in ADVANCED; also the DATE can be defaulted to the CURRENT DATE



Parent form

- Submitter is linked auto

FARM INFORMATION

FARM NAME

SURNAME

NAME

INITIALS

TELEPHONE

EMAIL

CELL

PRIVATE VET

DATE OF CENSUS

ADDRESS

Text

Numeric

Phone

Date

Time

Dropdown

Radio

Checkbox

Search

Text Box

Readme

Location

Photo

Audio

Video

Barcode

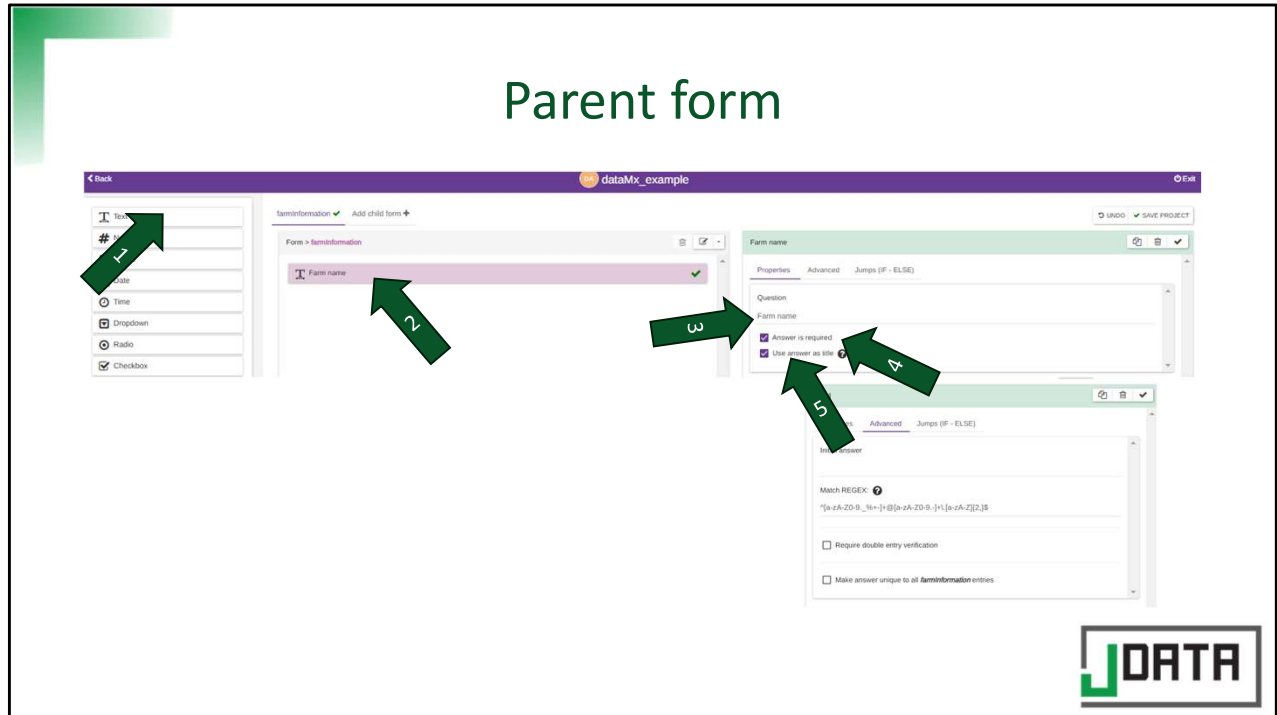
Branch

Group

JDATA

- Here we map each item in the paper form to a matching data type.

Parent form



- For farm name for instance – drag and drop a TEXT type from the left into the middle section
- In the settings section on the right indicate that the answer is required
- If an answer is USED AS TITLE it means the answer will form part of the identifier for that entry
- An initial answer can be given, generally for most data types – DATA STANDARDISATION

Parent form

The screenshot displays the JDATA software interface for configuring a 'Parent form'. On the left, a sidebar lists various input types: Text, Numeric, Phone, Date, Time, Dropdown, Radio, Checkbox, Search, Text Box, Readme, Location, Photo, and Audio. The main workspace shows a list of form fields for a form named 'farminformation'. Each field has a green checkmark indicating it is configured correctly. The fields are: Farm name, Surname, Name, Initials, Telephone, Email, Cellphone, Private vet (highlighted in pink), Census date, Farm location, and Address. A green arrow points to the 'Add child form +' button at the top of the form list. Another green arrow points to the 'Private vet' field. The right panel shows the 'Private vet' properties, including 'Initial answer' set to 'Unknown vet' and 'Match REGEX: ?'. The JDATA logo is in the bottom right corner.

- Here is an example of putting in 'UNKNOWN VET' as the initial answer
- After creating all inputs then click on ADD CHILD FORM to start with the second form of the investigation process

Regex and Double Entry

The image displays a form builder interface. On the left, a list of fields is shown with green checkmarks indicating they are configured correctly. The fields are: Farm name, Surname, Name, Initials, Telephone, Email (highlighted), Cellphone, Private vet, Certificate date, Farm location, and Address. On the right, the configuration for the 'Email' field is shown. It includes a 'Match REGEX' field with the value `*[a-zA-Z0-9_!@-]+@[a-zA-Z0-9-!]+[a-zA-Z]{2,6}` and a checkbox for 'Require double entry verification' which is checked. Below the main interface, there are two screenshots of the application in use. The left one is labeled 'Web application' and shows a 'Patient' form with an email field and a 'Confirm Answer' button. The right one is labeled 'Mobile application' and shows the same form on a mobile device, with the email field containing 'john.doe@gmail.com' and a 'Confirm answer here...' button.



- Here the email field has been allocated REGEX to ensure that the info is correctly captured
- Also a double entry is inserted to ensure that the user is definitely correct when capturing the data

Child form - equines

HORSE INFORMATION								
Name/ID	Passport/Chip	Sex	Age (if only year is known then choose 1 Aug of that year)	Colour	Breed	Species	Age (if only year is known then choose 1 Aug)	Age (if only year is known then choose 1 Aug)

Form name

equines

CLOSE SAVE CHANGES



- The CHILD FORM for the example
- ONE location can have MANY equines (like ONE parent can have MANY children)
- Again, each field in the paper form must be mapped to an appropriate type in the EpiCollect5 system
- Name the child form and SAVE CHANGES

The screenshot displays the 'dataMx_example' software interface. On the left is a sidebar with various data type icons: Text, Numeric, Phone, Date, Time, Dropdown, Radio, Checkbox, Search, Text Box, Readme, Location, Photo, Audio, Video, and Barcode. The central area shows a form titled 'Form > equines' with a list of fields, each with a green checkmark indicating it is mapped. The fields are: Equine name, Passport number, Microchip (highlighted with a green arrow and a barcode icon), Sex, Date of birth, Colour, Breed, Species, Last AHS 1 vaccination date, Last AHS 1 vaccination batch, Last AHS 2 vaccination date, Last AHS 2 vaccination batch, and Sample ID. On the right, the 'Equine name' question properties are shown, including options for 'Answer is required' and 'Use answer as title'.

- Here all fields are mapped – note the use of barcode for microchip

farmInformation ✓ equines ✓ Add child form +

Form > equines

Equine name	✓
Passport number	✓
Microchip	✓
Sex	✓
Date of birth	✓
Colour	✓
Breed	✓
Species	✓
Last AHS 1 vaccination date	✓
Last AHS 1 vaccination batch	✓
Last AHS 2 vaccination date	✓
Last AHS 2 vaccination batch	✓
Sample ID	✓

Sex

Properties Advanced Jumps (IF - ELSE)

Question

Sex

☐ Answer is required

☐ Use answer as title ?

Possible answers

+ ADD ANSWER

Unknown sex	x
Filly	x
Mare	x
Colt	x
Gelding	x

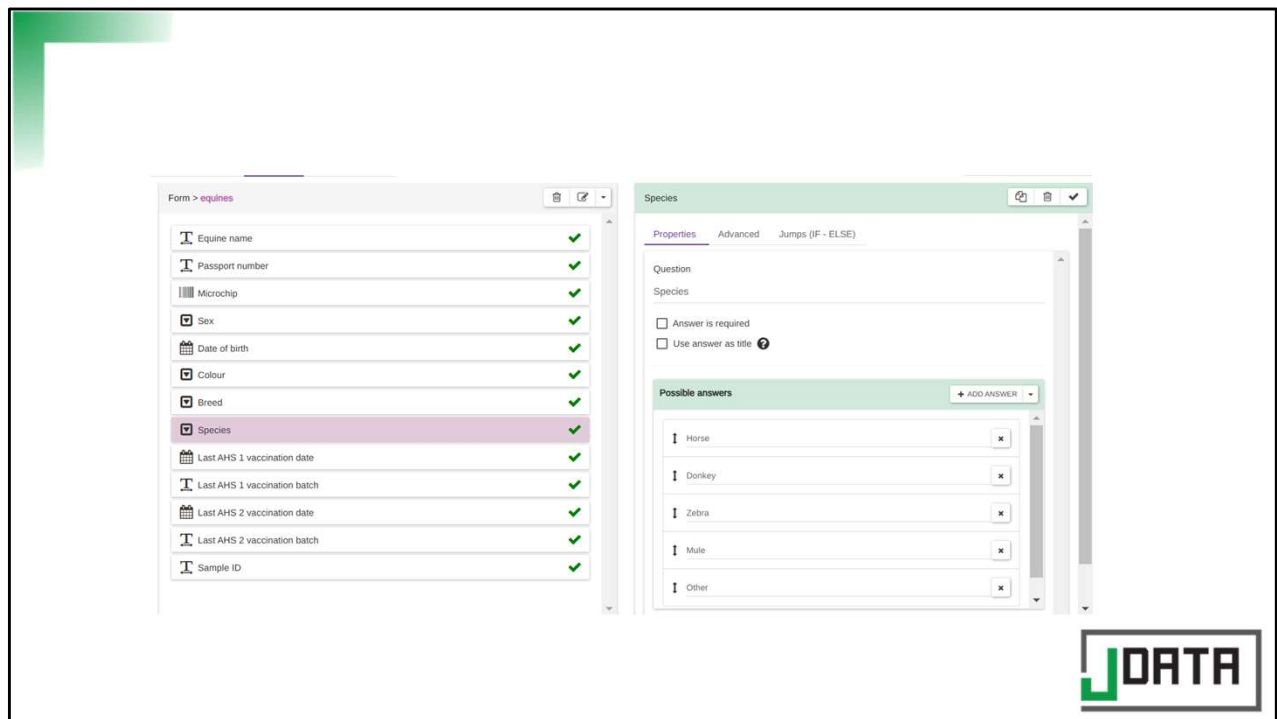
- Drop down options are crucial for standardisation of data. Here the SEX of the animal is categorised by common equine sex types but also an UNKNOWN type and general male and female classification

Importing options

The diagram illustrates the process of importing a list of options into a JDATA form. The workflow is as follows:

- Initial Form State:** A form titled "Colour" is shown in the "Properties" view. It has a "Question" field with the text "Colour" and a "Possible answers" section with a placeholder "I am a placeholder answer".
- Accessing Import Options:** The form is switched to the "Advanced" view. In the "Possible answers" section, the "Import CSV" button is clicked.
- Import Options Dialog:** A dialog box titled "Possible answers csv import options" appears. It allows the user to select a dataset to import (currently "colour.csv") and choose between "Find new content headers" and "Replace all existing possible answers".
- CSV Data Source:** A screenshot of a CSV file named "colour.csv" is shown. It contains a list of colors: Red, Orange, Yellow, Green, Blue, Purple, Brown, Grey, Black, White, Silver, Gold, Bronze, Copper, Steel, Iron, Lead, Zinc, Aluminium, Titanium, Stainless Steel, Plastic, Wood, Glass, Paper, Fabric, Leather, Rubber, Metal, and Unknown.
- Final Form State:** The form is switched back to the "Properties" view. The "Possible answers" section now displays the imported list of colors, with "Red" selected as the initial answer.

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- Same process for a dropdown like species – for short option lists simply type them in and move them into the order you prefer
- Always retain an unknown/undefined option

Saving

The screenshot shows the 'dataMx_example' application interface. On the left is a sidebar with various input types: Text, Numeric, Phone, Date, Time, Dropdown, Radio, Checkbox, Search, Text Box, Readme, Location, Photo, Audio, Video, Barcode, and Branch. The main area displays a form titled 'Form > equines' with fields for Equine name, Passport number, Microchip, Sex, Date of birth, Colour, Breed, Species, and several vaccination dates. Each field has a green checkmark indicating it is valid. On the right, the 'Species' dropdown menu is open, showing options: Horse, Donkey, Zebra, Mule, and Other. A green arrow points to the 'SAVE PROJECT' button in the top right corner of the form area.

⚠ A project can be saved only when ALL its questions are valid.



- Saving a project is only possible once all questions are validated
- A form (CHILD OR PARENT) cannot be saved unless at least one question exists
- Forms can be imported from previously saved form definitions

Privacy and Access

Visibility settings are primarily relevant to PUBLIC projects.
PRIVATE projects are already hidden from web searches on the Epicollect5 site.
On the mobile app, all projects are visible when searching. However, PRIVATE projects are marked with a "lock" icon, indicating that they require authentication for downloading. This ensures secure access to those projects.

epicollect5

Hi, John

My Projects

Create Project

Find Project

User Guide

Logout

dataMx_example

Homepage: https://five.epicollect.net/project/datamx-example

Dashboard

Details

Form Builder

Manage Users

Map Data

Clone

Manage Entries

Developers

API

Apps

Project details

DA

An example project for teaching purposes

Created on Wednesday 26 Feb 2025, 07:28

No description yet

Settings

Access

PRIVATE PUBLIC

Status

ACTIVE TRASH LOCK

Visibility

LISTED HIDDEN

Category

General

Found a bug? Have a question? Ask the Community at community.epicollect.net

Project Access

PRIVATE or PUBLIC, define whether the users need to be added to the project as members to view & collect entries.

- Public: anyone will be able to view the collected entries, add new entries, or download all the entries. API endpoints are open.
- Private: only members of the project can access it. API endpoints require authentication

Project Status

- ACTIVE: The project can be viewed, and data can be collected and uploaded.
- TRASH: The project cannot be viewed or receive data, but it can be restored. To delete it permanently, when a project is trashed a "delete" button appears.
- LOCK: The project can be viewed, but it does not accept any new entries.

Project Visibility

- LISTED: This setting makes your project publicly discoverable through the website's search feature.
- HIDDEN: When set to "HIDDEN," the project is accessible only to users who have the project's direct URL. This option provides increased privacy.



Project Details and Project Data

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Role	Description
CREATOR	A project creator originally created the project and has full access to the project, including viewing, editing, deleting and uploading data via the mobile client/web. A creator can add/remove any other type of users to the project, except for other creators.
MANAGER	A project manager has full access to the project, including viewing, editing, deleting and uploading data via the mobile client/web. A manager can add/remove curators and collectors to the project, but not managers. A manager can alter the project setting, even the forms.
CURATOR	A project curator has high access to the project, including viewing, editing and uploading data via the mobile client/web. A curator CANNOT alter the project settings or the forms. A curator cannot add other users to the project.
COLLECTOR	A project collector has basic access to the project, including viewing and uploading only their own data via the mobile client/web. This means COLLECTOR A cannot access entries uploaded by COLLECTOR B, and vice versa. A collector cannot make any changes to the project.
VIEWER	A project viewer gets READ ONLY access to a project. Viewers can view all the data collected by any other user but they cannot make any changes to the data or access any of the project settings.

A note on Project Users

	ADD CREATOR	ADD MANAGER	ADD CURATOR	ADD COLLECTOR
CREATOR	✗	●	●	●
MANAGER	✗	✗	●	●
CURATOR	✗	✗	✗	✗
COLLECTOR	✗	✗	✗	✗
VIEWER	✗	✗	✗	✗



- The User Management section in the Project details is where allocation of users to different levels is made.
- In private projects this is crucial to manage
- Users can be imported via importation

Data capture via the web link

The screenshot displays a web form interface with the following elements:

- Form Title:** dataMy_example
- Section 1: farminformation**
 - Field:** Farm name
 - Progress:** 0%
- Section 2: Email**
 - Field:** Email
 - Progress:** 45%
 - Error Message:** Regular expression format not matched.
- Section 3: Email**
 - Field:** Email
 - Text:** jdgrewar@gmail



- Data capture via the web will be demonstrated in person
- In this screenshot however an example is given of the format and also where an email address was entered incorrectly, preventing further capture of that data point.

- Location



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Data capture

100%

farmInformation

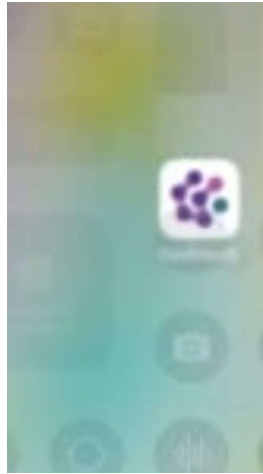
SAVE ENTRY

View	Delete	Edit	equines	Title	Created At	Farm name	Surname	Name	Initials	Telephone	Email	Cellphone	
					Test farm	26th Feb, 2025	Test farm	Grewar	John	JD	0218888888	jdgrewar@gmail.com	0836420610

- Once a row of data is captured on the web it is available directly – and here it can be edited/deleted/downloaded

Data capture - Mobile

<https://vimeo.com/1060440188?share=copy#t=0>



Consolidation

dataMx_example farminformation

Hi, John Logout Download Table Map Exit

Add farminformation Total: 2, 1/1

Filter by title FROM: 26 FEB. 25 TO: 26 FEB. 25 NEWEST X

View	Delete	Edit	equines	Title	Created At	Farm name	Surname	Name	Initials	Telephone	Email	Cellphone
			1	Test 2	26th Feb, 2025	Test 2	Clarborne	Dolores	D	123456789	Dolores@gmail.com	123456789
			1	Test farm	26th Feb, 2025	Test farm	Greear	John	JD	0218888888	jgreear@gmail.com	0836420610

← farminformation entries equines of "Test 2"

Add equines Total: 1, 1/1

Filter by title FROM: 26 FEB. 25 TO: 26 FEB. 25 NEWEST X

View	Delete	Edit	Title	Created At	Equine name	Passport number	Microchip	Sex	Date of birth	Colour	Breed	S
			Blue	26th Feb, 2025	Blue	Q12344	9780340980507 Can type	Filly	0402/2025	Bay	American Quarter Horse	0



After mobile data is uploaded it is then available in the data view in the online environment

Data can now still be edited and updated

Editing the data on mobile, with upload, will result in edits to the web data

dataMx_example farminformation

Hi, John Logout Download Table Map Exit

2, 1/1

Filter by title

FROM: 26 FEB, 2025 TO: 26 FEB, 25 NEWEST X

View	Delete	Edit	equines	Title	Created At	Farm name	Surname	Name	Initials	Telephone	Email	Cellphone
			1	Test 2	26th Feb, 2025	Test 2	Clairborne	Dolores	D	123456789	Dolores@gmail.com	123456789
			1	Test farm	26th Feb, 2025	Test farm	Grewar	John	JD	0218888888	jdgrewar@gmail.com	0836420610

Total: 2

JDATA

- Click on the MAP view to check the spatial distribution of the data

Accessing data

The screenshot shows the 'dataMx_example farminformation' interface. On the left, there's a sidebar with 'Download' and 'Mapping' sections. The 'Download' section has a 'DOWNLOAD' button. In the top right, there's a 'Download' button next to the user profile. A table of farm information is displayed in the center. Below the table, a ZIP file icon labeled 'datamx-example-csv.zip' is shown. To the right, a file explorer window shows the contents of the unzipped ZIP file, which are two CSV files: 'form-1_farminformation.csv' and 'form-2_equines.csv'. Green arrows numbered 1 through 4 indicate the sequence of actions: 1. Clicking the top right 'Download' button, 2. Clicking the sidebar 'DOWNLOAD' button, 3. The downloaded ZIP file, and 4. The unzipped CSV files.

Title	Created At	Farm name	Surname	Name	Initials	Telephone	Email	Cellphone
Test 2	26th Feb. 2025	Test 2	Clarborne	Dolores	D	123456789	Dolores@gmail.com	123456789
Test farm	26th Feb. 2025	Test farm	Grewar	John	JD	0218888888	jdgrewar@gmail.com	0836420610

- Data can easily be downloaded from EpiCollect by CSV
- Go to the Data view
- Click on DOWNLOAD, leave the mapping to default, leave the format to CSV and download
- Because in this case we have parent and child forms the data will be in a ZIP file
- Save this and unzip it on your computer, and review the two forms within

Data now available

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
ec5_uuid	created_at	uploaded_at	created_by	title	1_Farm_name	2_Surname	3_Name	4_Initials	5_Telephone	6_Email	7_Cellphone	8_Private_vet	9_Census_date	lat_10_Farm_location
6c3d6b42-05c8-42e8-af75-f9ce1fd13c85	2025-02-26T11:03:08.000Z	2025-02-26T11:05:19.000Z	jdgrewar@gmail.com	Test 2	Test 2	Clairborne	Dolores	D	123456789	Dolores@gmail.com	123456789	No Vet	26/02/2025	-34.414383
a79d0320-442f-11ef-85d0-07b613b93c99	2025-02-26T10:51:38.450Z	2025-02-26T10:54:21.000Z	jdgrewar@gmail.com	Test farm	Test farm	Grewar	John	JD	0218888888	jdgrewar@gmail.com	0836420610	Unknown vet	26/02/2025	-34.423889

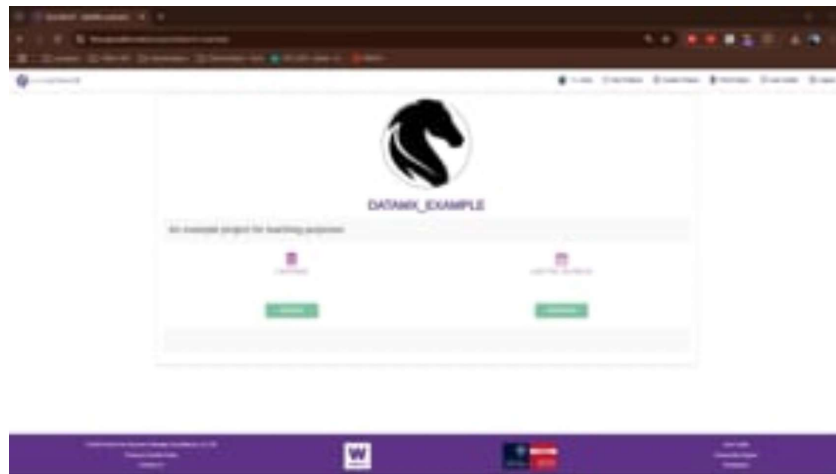
P	Q	R	S	T	U	V
long_10_Farm_location	accuracy_10_Farm_location	UTM_Northing_10_Farm_location	UTM_Easting_10_Farm_location	UTM_Zone_10_Farm_location	11_Address	
19.202543		4	6190431	334809 34H	Hermanus Beach Club	
19.196944		4	6189368	334313 34H	Test at Address	

A	B	C	D	E	F	G	H	I	J	K	L
ec5_uuid	ec5_parent_uuid	created_at	uploaded_at	created_by	title	12_Equine_name	13_Passport_number	14_Microchip	15_Sex	16_Date_of_birth	17_Colour
788b70b-1c2d-4462-a201-317a0d38cd5	6c3d6b42-05c8-42e8-af75-f9ce1fd13c85	2025-02-26T11:04:51.000Z	2025-02-26T11:05:21.000Z	jdgrewar@gmail.com	Blue	Blue	012344	9780340980507	Can type	Filly	04/02/2025
1cd20c30-4430-11ef-a41d-f76d4d046c72	a79d0320-442f-11ef-85d0-07b613b93c99	2025-02-26T10:54:55.091Z	2025-02-26T10:56:18.000Z	jdgrewar@gmail.com	Hamish	Hamish	No passport		9.8514E+16	Station	03/02/2010

M	N	O	P	Q	R	S
18_Breed	19_Species	20_Last_AHS_1_vaccin	21_Last_AHS_1_vaccin	22_Last_AHS_2_vaccin	23_Last_AHS_2_vaccin	24_Sample_ID
American Quarter Horse	Other	03/02/2025	123			None
xbreed	Mule	01/02/2025	123A	07/03/2025	123B	



Alternative mapping



Self-Paced exercise instructions

- Task 1: Review slide show and video material from the morning session
- Task 2:
 - Create and login into your EpiCollect5 environment
 - Create a data capture project that is private – you can choose any form you'd like or replicate the demo form
 - Name the Project anything you like but start the name with *DataMx_*
 - Add jdgrewar@gmail.com as a manager on your project to assist with follow up
 - Log some real or fake data using both the mobile and web-based form
 - Download the data into a CSV format
 - Bonus points for
 - Adding REGEX
 - Adding Jumps
 - Adding Groups
 - Adding Branches



Group discussions

- 15:00 Optional discussion slots
 - <https://us02web.zoom.us/j/85951489858?pwd=Gh0f2jaTwy0B1crY1Z2Z6g00bm1Xfl5.1>
Meeting ID: 859 5148 9858
Passcode: 011930
- 16:30 – 17:00 Wrap and discussion
 - <https://us02web.zoom.us/j/83981958564?pwd=IAH4SVRMx2bjm7yzWoQapaltBfjeMX.1>
Meeting ID: 839 8195 8564
Passcode: 603058
- Topics: Progress and technical challenges

